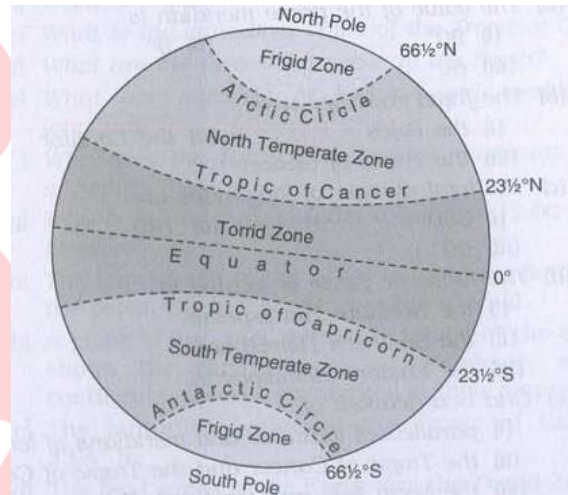


GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
GEOGRAPHY
CLIMATOLOGY
MODULE 1
EARTH'S COORDINATES, MOTIONS
CONCEPT OF TIME ZONES

LATITUDE

Latitude means how far a place is from the Equator, measured in degrees, towards the North or the South. Imagine lines drawn around the globe, going from left to right (like rings around an orange). These lines are called **parallels of latitude** because they run parallel to each other and never meet.

The **Equator** is the starting line. It is given the value **0 degrees**. It divides the Earth into two equal halves - the Northern Hemisphere and the Southern Hemisphere. From the Equator, latitude is counted upward to **90 degrees N (North Pole)** and downward to **90 degrees S (South Pole)**. So latitude values range from 0 degrees to 90 degrees North or South. Latitude is actually the angle formed at the centre of the Earth between the Equator and that particular place.



<u>Latitude Line</u>	<u>Value</u>
<u>Equator</u>	0 degrees
<u>Tropic of Cancer</u>	23.5 degrees N
<u>Tropic of Capricorn</u>	23.5 degrees S
<u>Arctic Circle</u>	66.5 degrees N
<u>Antarctic Circle</u>	66.5 degrees S
<u>North Pole</u>	90 degrees N
<u>South Pole</u>	90 degrees S

Countries passing through Equator : There are 13 countries - Ecuador, Colombia, Brazil, São Tomé and Príncipe, Gabon, Republic of the Congo, Democratic Republic of the Congo, Uganda, Kenya, Somalia, the Maldives (through its territorial waters), Indonesia, and Kiribati.

Countries passing through Tropic of Cancer : It passes through 17 sovereign countries - Mexico, Mauritania, Mali, Algeria, Niger, Libya, Egypt, Saudi Arabia,

United Arab Emirates, Oman, India, Bangladesh, Myanmar, and China, while also crossing the territorial waters of The Bahamas

Countries passing through Tropic of Capricorn : It passes through 10 sovereign countries - Chile, Argentina, Paraguay, Brazil, Namibia, Botswana, South Africa, Mozambique, Madagascar, and Australia.

All latitude lines are circles, but they are not all the same size. The Equator is the biggest circle. As you move towards the poles, the circles become smaller and smaller, until at the poles they shrink to a single point. Because the Equator is the only latitude that divides the Earth into two exactly equal halves, it is called a **Great Circle**. All other latitudes are called **Small Circles**.

Latitude also decides how much heat and sunlight a place gets. Places near the Equator get sunlight almost straight overhead all year, so they are hot. Places near the poles get sunlight at a slanting angle, spread over a larger area, so they are cold. This is why Earth is divided into heat zones:

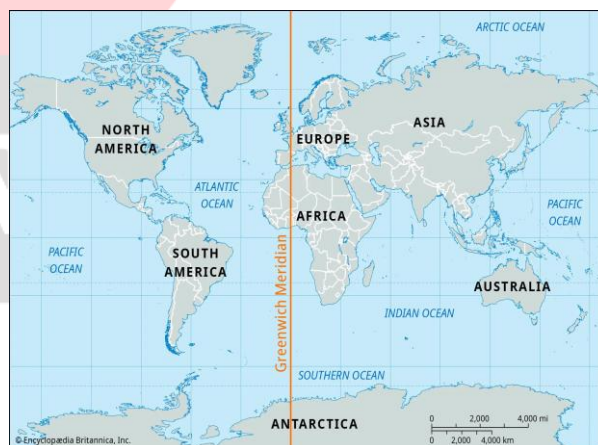
<u>Heat Zone</u>	<u>Temperature</u>	<u>Extent</u>
<u>Torrid Zone</u>	Hot	Between Tropic of Cancer and Tropic of Capricorn
<u>Temperate Zone</u>	Mild	Between the Tropics and the Polar Circles (two zones: North Temperate and South Temperate)
<u>Frigid Zone</u>	Cold	Between the Polar Circles and the Poles (two zones: North Frigid and South Frigid)

LONGITUDE

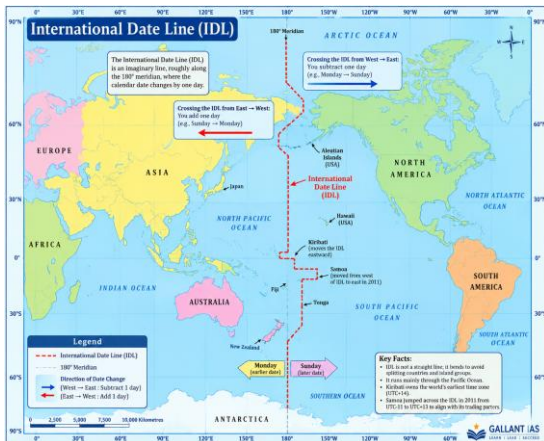
Longitude means how far a place is, in degrees, to the East or West of a fixed reference line. Imagine lines drawn from the North Pole straight down to the South Pole, like the segments of an orange. These lines are called **meridians of longitude**. Unlike latitude, there is no natural starting point like the Equator for longitude, so countries agreed on one by convention: the line passing through **Greenwich, London (UK)** is fixed as **0 degrees**, called the **Prime Meridian**. From the Prime Meridian, longitude is counted up to **180 degrees**

towards the East and **180 degrees** towards the West. So longitude values range from 0 degrees to 180 degrees E or 0 degrees to 180 degrees W.

Every longitude line is a **semi-circle** - it runs from Pole to Pole and then needs a second semi-circle on the opposite side to complete a full circle. Because of this, all



longitudes, when paired with their opposite one, form **Great Circles of equal size**. This is different from latitude, where only the Equator is a Great Circle. There are **360 meridians** in total (179 unique meridians East + 179 West + the Prime Meridian (0 degrees) + the 180 degree line = 360).



$360 \text{ degrees} / 24 \text{ hours} = 15 \text{ degrees of longitude} = 1 \text{ hour}$

$15 \text{ degrees} / 60 \text{ minutes} = 1 \text{ degree of longitude} = 4 \text{ minutes}$

The longitude exactly opposite to the Prime Meridian, at **180 degrees**, is called the **International Date Line (IDL)**. This is where the calendar date changes by one day. It is not a straight line - it zig-zags in places so that it doesn't cut through the middle of countries or island groups (otherwise the same country could have two different dates on either side of

a straight line).

MOTIONS OF THE EARTH

The Earth has two main motions happening at the same time.

1. Rotation (spinning on its own axis)

The Earth spins around an imaginary line passing through the North and South Poles, called its **axis**. This axis is not perfectly upright - it is **tilted at 23.5 degrees** from the vertical (perpendicular to Earth's orbital path). This tilt is extremely important and explains many things. One complete rotation takes about **24 hours** - this is what we call one day. The direction of rotation is **West to East**. This is why the Sun appears to rise in the East and set in the West (the Earth is turning towards the Sun in that direction, not the Sun actually moving around us).

Effects of rotation

- Causes **day and night**.
- Affects wind and ocean current directions (winds get deflected - this is called the **Coriolis effect** - to the right in the Northern Hemisphere and to the left in the Southern Hemisphere).
- Causes a **time difference** between places at different longitudes.

2. Revolution (moving around the Sun)

While spinning, the Earth also travels in a path (called an **orbit**) around the Sun. One complete revolution takes about **365 and a quarter days** - this is what we call one year. Because it is around 365 days AND 6 hours (precisely speaking it is 365 days 5h 48m, we round it to 6 hours for easy calculation), those extra 6 hours pile

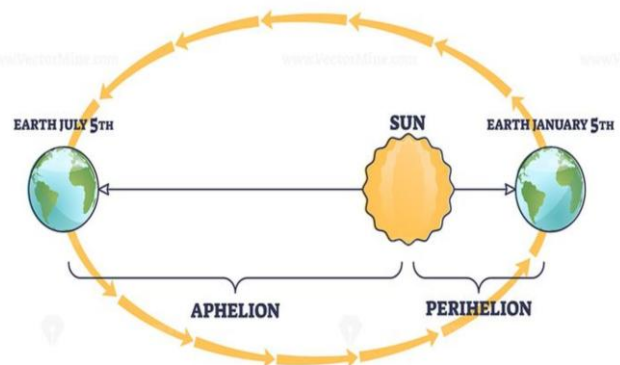
up. Every 4 years, they add up to almost one full extra day, which is added to the calendar as **29th February**. That year is called a **Leap Year**.

The Earth's path around the Sun is not a perfect circle but a slight oval (**ellipse**). So the distance between Earth and Sun keeps changing slightly through the year:

Perihelion: the point when Earth is closest to the Sun (around 3rd January).

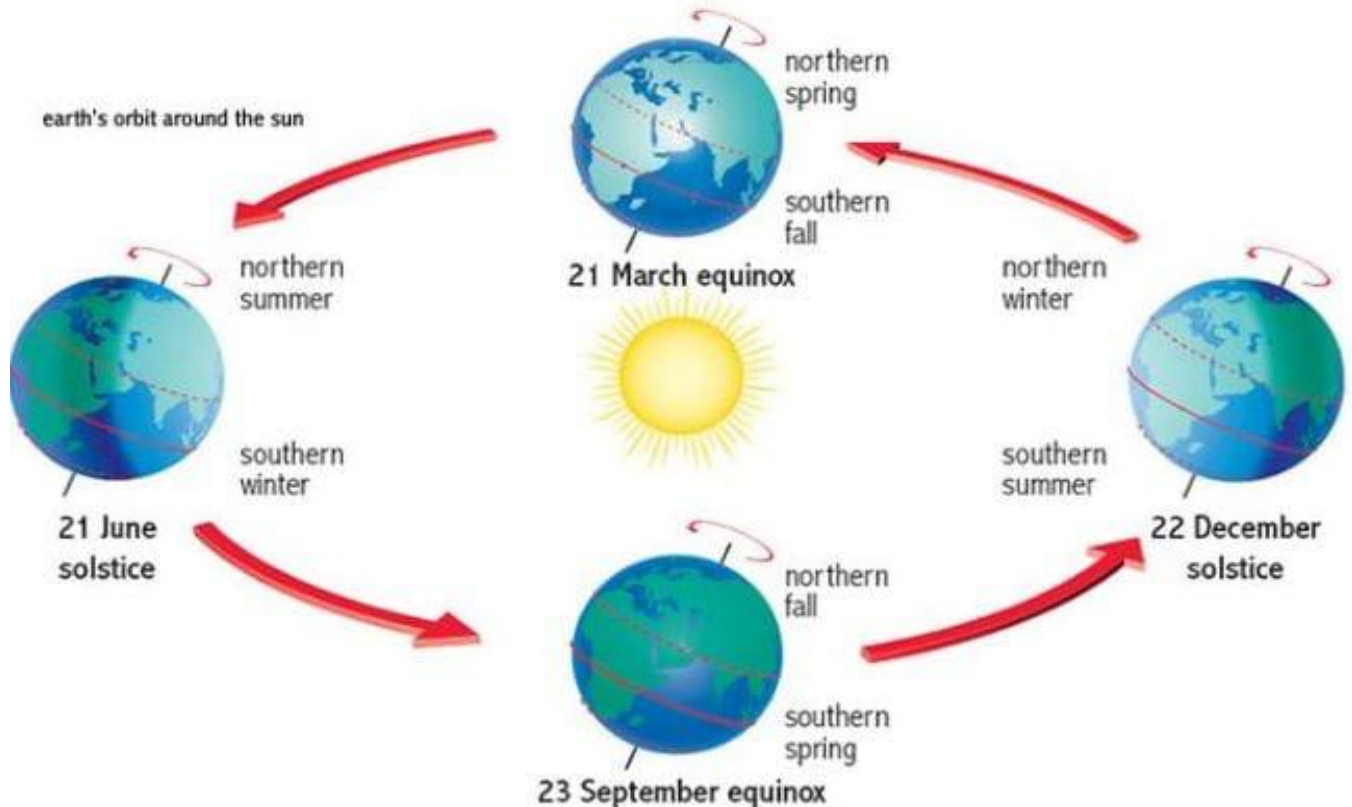
Aphelion: the point when Earth is farthest from the Sun (around 4th July).

PERIHELION AND APHELION



Why do we have seasons?

This is the most important concept to understand clearly. Because the axis is tilted at 23.5 degrees, as the Earth goes around the Sun, sometimes the Northern Hemisphere leans towards the Sun, and six months later, the Southern Hemisphere leans towards the Sun instead. The tilt itself never changes direction - it always points the same way in space - but Earth's position relative to the Sun changes as it orbits. When a hemisphere leans towards the Sun, it receives more direct, concentrated sunlight and longer days - so it experiences **summer**. The hemisphere leaning away gets slanted sunlight and shorter days - so it experiences **winter**.



<u>Date</u>	<u>Event</u>	<u>What Happens</u>
<u>20 or 21 June</u>	<u>Summer Solstice</u>	The Sun's rays fall directly (overhead) on the Tropic of Cancer (23.5 degrees N). The Northern Hemisphere has its longest day and shortest night. It is summer in the North and winter in the South.
<u>21 or 22 December</u>	<u>Winter Solstice</u>	The Sun's rays fall directly on the Tropic of Capricorn (23.5 degrees S). The Northern Hemisphere has its shortest day and longest night. It is winter in the North and summer in the South.
<u>20 or 21 March</u>	<u>Spring / Vernal Equinox</u>	The Sun's rays fall directly on the Equator. Day and night are equal (12 hours each) everywhere on Earth. ("Equinox" literally means "equal night".)
<u>22 or 23 September</u>	<u>Autumnal Equinox</u>	Again the Sun is directly over the Equator, and day and night are equal everywhere.

TIME

Time on Earth is measured based on the position of the Sun, which in turn depends on **longitude**.

Local Time

Local Time is the time calculated for a specific place, based on when the Sun is exactly overhead (local noon) for that place's own longitude. Since every degree of longitude equals 4 minutes, two places even a little apart in longitude will technically have slightly different local times. If every single town went by its own exact local time, train and flight schedules across a country would become extremely confusing, since even nearby towns would differ by a few minutes. To solve this, countries adopt one single time for the whole nation (or region), called **Standard**

Time. (In some cases a country may have multiple official time zones example - Russia)

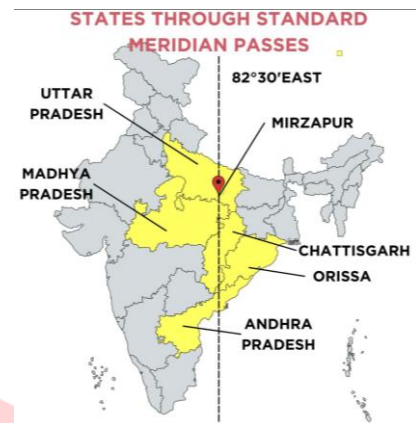
Standard Time

A country selects one central meridian (longitude) running through it, calculates the local time at that meridian, and applies that same time to the entire country, regardless of each place's own exact longitude.

Indian Standard Time (IST)

India's standard meridian is **82.5 degrees E**, which passes through **Mirzapur**, near Prayagraj (Allahabad), Uttar Pradesh. This longitude was chosen because it roughly passes through the middle of the country. IST is **5 hours 30 minutes ahead of GMT** (Greenwich Mean Time), written as GMT +5:30.

India uses a single time zone for the entire country, even though India's actual longitudinal spread (from about 68 degrees E to 97 degrees E) would normally cover more than one time zone. This means places in the far east of India (like Arunachal Pradesh) see the sun rise much earlier than their clock "should" suggest compared to places in the west (like Gujarat).



Does India require multiple time zones?

For multiple time zones:

1. **Sunlight timing** - places like Arunachal Pradesh get daylight almost 2 hours before Gujarat, but both follow the same clock.
2. **Saves energy** - matching clock time to daylight means less use of lights and electricity in the morning/evening.
3. **Better for health** - the body clock works better when clock time matches actual sunrise and sunset.
4. **Fairer timings** - children in the east often go to school while it's still dark; people in the west may work later into the evening.

Against multiple time zones:

1. **Transport confusion** - trains, flights, and buses crossing time zones would need constant time changes, making schedules harder.
2. **National unity** - one common time helps people across the country feel connected and simplifies daily coordination.
3. **Business complications** - banks, stock markets, and offices working across states would need to manage multiple clocks.
4. **Governance and safety** - coordinating government work, defence, and emergency services becomes harder with different times in different states.

Factors like impact of sunrise and sunset timings on biological activities of people, synchronizing sunrise and sunset timings with office hours should be carefully studied. It is important to consult all stakeholders and arrive at a better solution so as to use the lost light hours that are dimmed by a common IST.

GMT (Greenwich Mean Time) / UTC

The time at the Prime Meridian (0 degrees) is used as the world's reference point. All other time zones are described as being a certain number of hours "ahead of" or "behind" GMT.

Since 1 degree = 4 minutes, to find the time difference between two places, find the difference in their longitudes (in degrees), then multiply by 4 to get minutes. Places to the **East** are **ahead** in time (they see the sunrise earlier); places to the **West** are **behind** in time.

Example: If it is 12:00 noon at the Prime Meridian (0 degrees), the time at 82.5 degrees E would be $12:00 + (82.5 \times 4 \text{ minutes}) = 12:00 + 330 \text{ minutes} = 12:00 + 5 \text{ hrs } 30 \text{ min} = \underline{\underline{5:30 \text{ PM}}}$.

PREVIOUS YEAR QUESTIONS

1. Consider the following statements: [2025]
 - I. Anadyr in Siberia and Nome in Alaska are a few kilometers from each other, but when people are waking up and getting set for breakfast in these cities, it would be different days.
 - II. When it is Monday in Anadyr, it is Tuesday in Nome.Which of the statements given above is/are correct?
 - (a) I only
 - (b) II only
 - (c) Both I and II
 - (d) Neither I nor II
2. Consider the following countries [2024]
 1. United Kingdom
 2. Denmark
 3. New Zealand
 4. Australia
 5. BrazilHow many of the above countries have more than four time zones?
 - A) All five
 - B) Only four
 - C) Only three
 - D) Only two
3. On June 21 every year, which of the following latitude(s) experience(s) a sunlight of more than 12 hours? [2024]
 1. Equator
 2. Tropic of Cancer
 3. Tropic of Capricorn
 4. Arctic CircleSelect the correct answer using the code given below:
 - (a) 1 only
 - (b) 2 only
 - (c) 3 and 4
 - (d) 2 and 4

4. Variations in the length of daytime and night-time from season to season are due to: [2013]
- (a) the earth's rotation on its axis
 - (b) the earth's revolution round the sun in an elliptical manner
 - (c) latitudinal position of the place
 - (d) revolution of the earth on a tilted axis
5. In the northern hemisphere, the longest day of the year normally occurs in the: [2022]
- (a) First half of the month of June
 - (b) Second half of the month of June
 - (c) First half of the month of July
 - (d) Second half of the month of July
6. On 21st June, the Sun [2019]
- (a) does not set below the horizon at the Arctic Circle
 - (b) does not set below the horizon at Antarctic Circle
 - (c) shines vertically overhead at noon on the Equator
 - (d) shines vertically overhead at the Tropic of Capricorn
7. Consider the following countries [2009]
- 1. Australia
 - 2. Namibia
 - 3. Brazil
 - 4. Chile
- Through which of the above does the Tropic of Capricorn pass ?
- (a) 1 only
 - (b) 2, 3 and 4
 - (c) 1, 2 and 3
 - (d) 1, 2, 3 and 4
8. Which one of the following straits is nearest to the International Date Line? [2008]
- (a) Malacca Strait
 - (b) Bering Strait
 - (c) Strait of Florida
 - (d) Strait of Gibraltar
9. Which one of the following cities does NOT have the same clock time as that of the other cities at any given instant? [2007]
- (a) London (UK)
 - (b) Lisbon (Portugal)
 - (c) Accra (Ghana)
 - (d) Addis Ababa (Ethiopia)
10. Among the following cities, which one is nearest to the Tropic of Cancer? [2003]
- (a) Delhi
 - (b) Kolkata
 - (c) Jodhpur
 - (d) Nagpur

11. Which one of the following meridians did India experience the first light of the sunrise of the new millennium? [1999]
(a) $2^{\circ} 30' W$
(b) $82^{\circ} 30' E$
(c) $92^{\circ} 30' W$
(d) $92^{\circ} 30' E$
12. If it is 10.00 a.m. I.S.T., then what would be the local time at Shillong on $92^{\circ} E$ longitude? [1999]
(a) 9.38 a.m.
(b) 10.38 a.m.
(c) 10.22 a.m.
(d) 9.22 a.m.
13. A ship sailing from the eastern extremity of the Aleutian Islands to Dutch Harbour crosses the 180° meridian at 23.30 hrs on January 1, 1999. What time and date will be recorded by the captain of the ship in his diary after a one hour journey from the point of crossing of the meridian? [1999]
(a) January 1, 0030 hrs
(b) January 2, 0030 hrs
(c) January 3, 0030 hrs
(d) January 4, 0030 hrs
14. When there is noon at the G.M.T. meridian, people at another place of the Earth are taking their 6 o'clock morning tea. The longitude of the place is [1998]
(a) $17^{\circ} 30' E$
(b) $7^{\circ} 30' W$
(c) $172^{\circ} 30' E$
(d) $90^{\circ} W$
15. If the earth's direction of rotation is reversed, what would be the IST when it is noon at the International Date Line? [1997]
(a) 06.30 hrs
(b) 05.30 hrs
(c) 18.30 hrs
(d) 17.30 hrs

NCERT Reference

Old NCERT - Class 6th - Chapter 2 & 3 (or New NCERT Class 6 Chapter 1)

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